EXP.NO: 4	**Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.**

AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. Input:

- The user fills out a web form with their **name** and **age** and submits it.

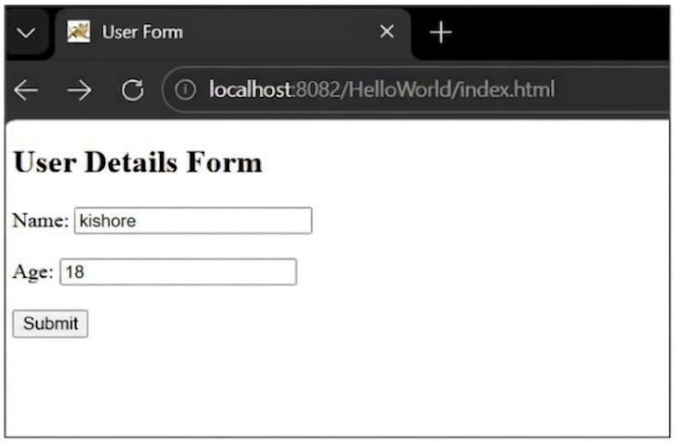
2. Form Handling and Servlet Processing:

1. **Create the Web Form:**
    - Create an HTML form with **two input fields**: one for the user's **name** and one for the user's **age**.
    - Add a **submit button** to the form.
    - Set the form's action attribute to the URL of the **Servlet** and the method to POST (or GET).
2. **Create the Servlet Class:**
    - Write a Servlet that processes the **form data** submitted by the user.
    - The Servlet will override the doPost() (or doGet() depending on the form method) to handle form submissions.
    - The Servlet will **retrieve the name and age** parameters from the request object using request.getParameter("name") and request.getParameter("age").
3. **Process the Data in the Servlet:**
    - Extract the name and age values submitted from the form.
    - **Validate the input** (e.g., check if the age is a valid number).
    - Prepare a **response** that will display the user's **name** and **age** back on the browser.
4. **Send Response to the Browser:**
    - The Servlet will send a **response** to the browser containing a message like:

        - "Hello, [name]! You are [age] years old."
    - The response will be displayed to the user in the browser.

5. **Deploy the Servlet:**
    - Compile the Servlet and deploy it to a **Servlet container** (like **Apache Tomcat**).
    - Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the form's action attribute.
6. **Test the Application:**
    - Open the web browser, fill in the form with the name and age, and submit it.
    - The Servlet will process the data and display the message back to the user in the browser.

Output:



HTML File (index.html):

```
<!DOCTYPE html>

<html>
```

```
<head>
    <title>User Form</title>
</head>
<body>
    <h2>User Details Form</h2>
    <form action="UserServlet" method="post">
        Name: <input type="text" name="name" required><br><br>
        Age: <input type="text" name="age" required><br><br>
        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

Java File (UserServlet.java) :

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

public class UserServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
```

```
        String name = request.getParameter("name");
        String ageStr = request.getParameter("age");

        try {
            int age = Integer.parseInt(ageStr);

            out.println("<html><body>");
            out.println("<h2>Hello, " + name +
                    "! You are " + age + " years old.</h2>");
            out.println("</body></html>");

        } catch (NumberFormatException e) {
            out.println("<html><body>");
            out.println("<h2>Invalid age entered.</h2>");
            out.println("</body></html>");
        }
    }
}
```

Java Compiled File (UserServlet.class) :

```
// Source code is decompiled from a .class file using FernFlower decompiler (from Intellij
IDEA).
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;
```

```
public class UserServlet extends HttpServlet {
  public UserServlet() {
  }

  protected void doPost(HttpServletRequest var1, HttpServletResponse var2) throws
ServletException, IOException {
    var2.setContentType("text/html");
    PrintWriter var3 = var2.getWriter();
    String var4 = var1.getParameter("name");
    String var5 = var1.getParameter("age");

    try {
      int var6 = Integer.parseInt(var5);
      var3.println("<html><body>");
      var3.println("<h2>Hello, " + var4 + "! You are " + var6 + " years old.</h2>");
      var3.println("</body></html>");
    } catch (NumberFormatException var7) {
      var3.println("<html><body>");
      var3.println("<h2>Invalid age entered.</h2>");
      var3.println("</body></html>");
    }

  }
}
```

File (web.xml) :

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
        xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
        https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
        version="6.0">

    <servlet>
        <servlet-name>HelloWorld</servlet-name>
        <servlet-class>HelloWorldServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>HelloWorld</servlet-name>
        <url-pattern>/hello</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>UserServlet</servlet-name>
        <servlet-class>UserServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>UserServlet</servlet-name>
        <url-pattern>/UserServlet</url-pattern>
    </servlet-mapping>

</web-app>
```

RESULT:

Thus A Servlet was created to process the HTML form data and display it back on the browser.